

Erwin Quanten

Staff Software Engineer – Python, Django, FastAPI, Angular, AWS

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PROFESSIONAL SUMMARY

Staff Software Engineer with 16+ years of experience building secure **Python 3.8–3.12** and **Angular 8–17** platforms across healthcare, logistics, energy, SaaS, and manufacturing industries in one full-stack delivery role. Skilled in **FastAPI, Django, Flask, TypeScript, RxJS, NgRx, PostgreSQL, AWS**, Docker, Kubernetes, and CI/CD, with strong ownership of legacy migration, API modernization, performance tuning, and cloud deployment. Known for resolving production defects, improving release reliability, reducing operational risk, and delivering scalable enterprise features with practical architecture, clean collaboration, and measurable business impact.

SKILLS

- **Backend:** **Python 3.8–3.12, FastAPI 0.95–0.110, Django 2.2–5.0**, Flask, Celery, SQLAlchemy, REST API, GraphQL API
- **Frontend:** **Angular 8–17, TypeScript 3.7–5.x**, RxJS, NgRx, Angular Material, Reactive Forms, Micro Frontends
- **Cloud & DevOps:** **AWS**, ECS, EKS, Lambda, S3, RDS, CloudWatch, Docker, Kubernetes, Terraform, GitHub Actions, Jenkins
- **Database:** **PostgreSQL**, MySQL, Redis, DynamoDB, query optimization, indexing, migration scripting
- **Architecture:** Microservices, event-driven systems, API gateway, domain-driven design, authentication, RBAC, observability
- **Testing:** Pytest, Jasmine, Karma, Cypress, Playwright, contract testing, integration testing

EXPERIENCE

Lightedge — Lead Software Engineer

Remote/ Full time

May 2022 – Apr 2026

Houston, TX

- Led enterprise cloud platform development in the infrastructure and managed-services industry using **Python 3.10–3.12, FastAPI, Angular 14–17, TypeScript, AWS, PostgreSQL**, Docker, and Kubernetes.
- Designed high-throughput **FastAPI** services with async workers, Redis caching, and PostgreSQL indexing, improving API response time by **28%** under customer portal load.
- Migrated legacy Angular modules from **Angular 11** patterns to **Angular 16–17** standalone components, typed forms, and optimized lazy loading, reducing bundle size by **22%**.
- Resolved a recurring production issue where expired JWT refresh tokens caused silent logout loops, adding interceptor-level retry handling and stronger backend token validation.
- Built role-based customer dashboard features with **Angular Material, NgRx**, and secure API contracts, giving enterprise clients clearer visibility into billing, incidents, and infrastructure health.
- Improved CI/CD reliability by separating test, build, image scanning, and deployment stages in GitHub Actions, reducing failed releases by **31%**.
- Implemented structured logging, correlation IDs, and CloudWatch dashboards for Python microservices, helping support engineers isolate customer-impacting incidents faster.
- Reworked a slow reporting endpoint by replacing nested ORM calls with optimized SQLAlchemy queries, reducing database load by **26%** during monthly reporting cycles.
- Introduced Pytest integration coverage for authentication, invoice export, and customer provisioning flows, preventing repeated regression bugs during release windows.
- Coordinated backend, frontend, QA, and DevOps teams through architecture reviews, release planning, and incident follow-ups without blocking parallel feature delivery.
- Implemented feature flags for high-risk Angular workflows, allowing gradual rollout of portal changes and quick rollback when customer-specific edge cases appeared.
- Mentored engineers on **Python, Angular**, clean API design, and production debugging, improving code review quality and reducing repeated implementation mistakes.

Syscom Digital — Staff Software Engineer

Remote/ Full time

Nov 2017 – Apr 2022

Bucharest, Romania

- Delivered full-stack SaaS and logistics platforms using **Python 3.7–3.10, Django 2.2–4.0, Flask, Angular 8–13, RxJS, PostgreSQL**, Celery, Docker, and AWS.

- Modernized a monolithic Django application into service-oriented modules with clearer domain boundaries, improving release flexibility by **24%** across several product teams.
- Migrated frontend applications from **Angular 6–7** to **Angular 12–13**, replacing outdated lifecycle patterns with typed services, reactive forms, and stronger component contracts.
- Solved a production shipment-status bug caused by duplicate Celery retry execution, adding idempotency keys and transactional updates to protect customer order accuracy.
- Built reusable Angular modules for search, filtering, authorization, and activity timelines, reducing duplicated frontend implementation by **29%** across internal dashboards.
- Optimized PostgreSQL queries for route planning and invoice reconciliation, cutting slow query volume by **21%** through indexing, pagination, and query plan review.
- Implemented secure REST APIs with Django REST Framework, JWT authentication, permission classes, and audit logging for partner-facing logistics workflows.
- Improved Docker-based local development by standardizing environment variables, seed data, and migration scripts so new engineers could run services with fewer manual steps.
- Added Cypress and Pytest coverage around critical booking, billing, and reporting flows, catching integration defects before staging deployments.
- Collaborated with product owners and distributed engineering teams to convert ambiguous logistics requirements into maintainable backend APIs and Angular user experiences.

Tessenderlo Group — Senior Software Engineer

Remote/ Full time

Aug 2015 – Oct 2017

Brussel Metropolitan Area

- Built manufacturing and supply-chain applications using **Python 3.5–3.6**, **Django 1.8–1.11**, **Flask**, **Angular 2–4**, **TypeScript**, PostgreSQL, RabbitMQ, and Linux services.
- Migrated selected AngularJS screens to early **Angular 2–4** components, improving maintainability by **18%** while keeping business-critical workflows available during transition.
- Developed Python APIs for plant inventory, batch tracking, and transport coordination, reducing manual spreadsheet reconciliation by **23%** for operational teams.
- Resolved a production data mismatch where timezone handling caused incorrect shipment cut-off calculations, normalizing UTC storage and local display formatting.
- Implemented RabbitMQ-based background processing for long-running export jobs, preventing browser timeouts and improving user completion rate by **17%**.
- Refactored Django models and service layers to separate production planning rules from database access, making future feature changes safer and easier to test.
- Created Angular forms for batch approval and shipment updates with validation rules, error states, and clear user feedback for factory operations staff.
- Improved application monitoring by adding structured logs around failed imports, delayed jobs, and API validation errors, reducing investigation time during incidents.
- Supported gradual migration from older server-rendered screens into API-driven Angular views while preserving user workflows and minimizing training disruption.

Tessenderlo Group — Software Engineer

Hybrid/ Full time

Sep 2011 – Jul 2015

Brussel Metropolitan Area

- Developed internal manufacturing and finance systems using **Python 2.7–3.4**, **Django 1.4–1.8**, **Flask**, **AngularJS 1.2–1.4**, jQuery, PostgreSQL, and Linux.
- Implemented AngularJS-based dashboards for procurement, inventory, and approval workflows, improving user task completion speed by **19%** compared with older page refresh screens.
- Migrated Python services from mixed **Python 2.7** scripts toward **Python 3.4** compatible modules, reducing future upgrade risk and improving code consistency.
- Fixed a recurring import failure where malformed CSV files broke nightly jobs, adding schema validation, reject-file reporting, and safer transaction handling.
- Built Django admin extensions and custom reporting views that reduced manual finance reconciliation effort by **16%** for monthly operational reporting.
- Improved PostgreSQL performance by adding targeted indexes and removing repeated query loops, reducing report generation time by **24%** on large datasets.
- Integrated authentication and permission checks across internal tools so factory, finance, and logistics users only accessed role-appropriate workflows.
- Worked flexibly across backend fixes, AngularJS UI changes, database scripts, and deployment support whenever production priorities required quick ownership.

Tessengerlo Group — Software Developer

Hybrid/ Full time

Jun 2009 – Aug 2011

Brussel Metropolitan Area

- Built early internal business applications using **Python 2.6–2.7**, **Django 1.1–1.3**, PostgreSQL, jQuery, HTML, CSS, and Linux deployment scripts.
- Automated repetitive reporting and data-cleaning scripts with Python, reducing manual operations effort by **14%** for finance and production support teams.
- Fixed database consistency issues caused by partial imports, adding transaction handling and validation checks that reduced repeated data correction work by **12%**.
- Created maintainable Django views, templates, and admin workflows for internal users, preparing backend APIs and data models for later AngularJS modernization.
- Supported production releases, bug triage, and user feedback loops with senior engineers, building strong habits around debugging, documentation, and careful change control.

EDUCATION

KU Leuven — Master's Degree, Computer Science

Sep 2003 – May 2009